

KNN → K nearest neighbours
 ↳ Objects that are close to each other are likely to be similar

Social proximity based decision making

→ 1. Classification
 know what we majority voting

→ 2. Regression
 average value of the nearest neighbours.

K → number of nearest neighbours.
 K=1, K=3

Student	Study Hours	Passed?	Distance G1	Distance
A	2	No	$\sqrt{(5-2)^2}$	3
B	3	No	$\sqrt{(5-3)^2}$	2
C	4	No	$\sqrt{(5-4)^2}$	1
D	6	Yes	$\sqrt{(5-6)^2}$	1
E	7	Yes	$\sqrt{(5-7)^2}$	2
F	8	Yes	$\sqrt{(5-8)^2}$	3

Q1. new student studied for 5 hours, will the student pass?
 Suppose K=3

Euclidean distance $\Rightarrow d = \sqrt{(x_1-x_2)^2 + (y_1-y_2)^2}$

ans:

Stud	Hours	Passed?
C	1	No
D	2	Yes
B	3	No
E	4	Yes
A	5	No
F	6	Yes

add. for 5
 K=3
 { 2 → No
 1 → Yes }
 so for 5 → No

1. $d = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$

2. Manhattan distance
 $d = \sum_{i=1}^n |x_i - y_i|$

3. Minkowski distance
 $d = \left(\sum_{i=1}^n |x_i - y_i|^p \right)^{1/p}$
 p → type of distance.
 p → 1
 p → 2

4. Hamming distance.

5. Cosine similarity

Data Type	Recommended
Continuous numeric data	Euclidean
grid-based movement	Manhattan
flexible generic case	Minkowski
binary or categorical	Hamming
text & high dimensional vectors	Cosine similarity.

K=1 → accuracy → 85%
 K=3 → → 89%
 K=7 → → 91%

K fold cross validation.

$K \approx \sqrt{n}$ → number of training samples.

25 → K=5
 100 → K=10

Statistical point.

K=7
 K=3
 K=11
 K=9

Curse of Dimensionality.

feature selection
 PCA
 F.S.
 Stand or Normalization.

effectiveness of them

$TPR = \frac{TP}{TP+FN}$

↓

Out of all positive cases how many did the model correctly identify.

$FPR = \frac{FP}{FP+TN}$

↓

Out of all negative cases how many were incorrectly classified as positive.

100 → correct patients.
 model predicts → 90
 TPR → 90%.

200 healthy patients.
 model wrong → 40
 $FPR = \frac{40}{200} = 20\%$

0.5
 0.1

Threshold → 1, 0.9, 0.8, 0.7, 0.6, 0.5, 0.4, 0.3, 0.2, 0.1, 0

TPR & FPR

one point.

Roc curve → X-axis → FPR
 Y-axis → TPR

AUC

1 → Perfect classifier

0.95 → Excellent.

0.90 → Very Good

0.80 → Good

0.5 → Random guessing

< 0.5 → worse.